Charles Kwanin

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Experience

June 2024 – present **Software Engineer II — Survey Taking Platform**, Qualtrics – Provo, UT

Backend services for survey response collection, serving hundreds of millions of requests per day across commercial and FedRAMP environments.

- Designed and launched a public survey-sessions API that opened a new revenue stream and an employee-engagement survey channel, cut p99 latency 44% under load, and now serves external customers and 5+ internal teams whose migrations I led.
- Unblocked Fortune 500 customers' move to the new survey-taking experience by bringing duplicate-respondent detection, reCAPTCHA v3 bot scoring, and honeypot fields to it behind one server-validated check.
- Led the Nomad-to-Kubernetes migration for multiple teams across the survey collection org, in every datacenter including FedRAMP/GovCloud, replacing bash-script deploys with Spinnaker and Jenkins with GitLab CI; rightsizing and autoscaling cut infrastructure spend.
- Built Go health probes for Kubernetes services, since adopted by every Go team on Kubernetes company-wide.
- Architected the public ingress layer for the survey-taking services, cutting NAT Gateway costs and exposure by taking service-to-service calls off the public internet, and isolating noisy-neighbor traffic with tiered per-survey and per-customer rate limits and a kill-switch.
- Contributed to the north-star architecture for the survey collection org, mapping current pain points and future direction; drove org-wide design reviews.
- Mentored 3 engineers, including an intern whose project I scoped and designed; resolved high-severity incidents and ran a Kubernetes game day to drill the team on incident response.

Sept 2022 – June 2024 **Software Engineer II — Survey Quality Platform**, Qualtrics – Provo, UT

Survey quality and approval platform covering question libraries, survey versioning, approval workflows, and quality scoring.

- Led a question-recommendation feature from design through launch, partnering with a data platform team on datastore selection and data modeling and mentoring a teammate through the build.
- Built a cross-service approval workflow and permissions model so employee-experience programs go live automatically once their survey is approved and published.
- Shipped core API routes and UI components for a redesigned question library, launched on schedule with 100% test coverage of critical integration flows.
- Owned and operated three production services, clearing every critical vulnerability in one through an OWASP ZAP audit.
- Expanded the survey quality checker to evaluate layout and question accessibility, automatically flagging non-WCAG 2.1 AA issues before launch.

July 2020 – Aug 2022 **Technology Analyst**, Goldman Sachs – Salt Lake City, UT

- Scaled event processing to tens of thousands of requests per minute by migrating a legacy proprietary framework to Java Spring Boot, Scala, Kafka, Elasticsearch, and MongoDB.
- Improved alerting accuracy and system diagnostic speed by instrumenting Prometheus metrics and Grafana dashboards across the fleet.
- Mentored 2 junior engineers and 2 interns, accelerating onboarding and reinforcing code review standards.

Education

Sept 2016 – May 2020 **Oberlin College**, BA in Computer Science – Oberlin, OH

Skills

Languages: Go, TypeScript, JavaScript, PHP, SQL, Java

Data and Messaging: MySQL, DynamoDB, MongoDB, Redis/Valkey, Kafka, SQS, Zod, data modeling & schema design

Cloud and Infrastructure: AWS (EC2, ELB, S3, SQS), Terraform, Kubernetes, Docker, Nomad, Spinnaker, Akamai, DNS

APIs and Testing: OpenAPI, Playwright, Cypress, OWASP ZAP, Twistlock, GitLab CI, Jenkins

Observability: Grafana, Prometheus, OpenTelemetry, Splunk